

How to Print PA6-CF (Nylon 6 Carbon Fiber) Filament

- Printer Requirements:
 - Use a hardened steel or ruby-tipped nozzle to withstand carbon fiber abrasiveness.
 - An enclosed printer is recommended to prevent warping.
 - Use a direct drive extruder for better control, though high-quality Bowden setups can also work.
- Recommended Print Settings:
 - Nozzle Temperature: 260–290°C (Check manufacturer specs)
 - Bed Temperature: 70–110°C
 - Layer Height: 0.4 mm or greater (due to fiber size)
- Print Surface & Adhesion:
 - Use PEI.
 - Apply a glue stick or a PA-specific adhesive to prevent sticking too hard or warping.
- Drying the Filament:
 - Critical step – Nylon is extremely hygroscopic.
 - Dry filament at 80-90°C for 12 hours before printing.
 - Check for <20% relative humidity before printing.
 - Store in a dry box or vacuum-sealed bag with desiccant.
- Speed & Retraction:
 - Print Speed: 60 mm/s (slower for better quality)
 - Retraction: 1–2 mm @ 20–40 mm/s (tune for your setup to avoid stringing)
- Notes
 - The body shrinks enough to warrant printing at 1.01-1.02 scale.
 - For a looser battery cage fit, scale the print 0.98 - 0.99.

How to Print PPA-CF (Carbon Fiber Reinforced Polythiazide) Filament

- Printer Requirements:
 - All-metal hotend rated for 320 °C or higher.
 - Hardened steel nozzle required (carbon fiber is abrasive).
 - Nozzle size: 0.4 mm+ (0.6 mm preferred to reduce clogging).
 - Heated enclosure recommended to prevent warping.
- Recommended Print Settings:
 - Nozzle Temperature: 280–320 °C (check your specific brand).
 - Bed Temperature: 80–120 °C (with adhesive aid).
 - Layer Height: 0.1–0.25 mm (0.2 mm typical).
 - Cooling Fan: Off or very low (0–30%).
- Print Surface & Adhesion:
 - Works well on PEI or Garolite with glue stick, Magigoo PA, or PVA glue.
 - Use a brim or raft for better first-layer grip.
 - Keep bed clean and pre-heated before printing.
- Drying the Filament:
 - PPA-CF is very hygroscopic; print only when dry.
 - Dry at 100–130 °C for 8–12 h (or 70 °C for 18–24 h in a filament dryer).
 - Store in a dry box or airtight container with desiccant.
 - Aim for <15% relative humidity.
- Speed & Retraction:
 - Print Speed: 30–80 mm/s (slower for fine detail).
 - Retraction: 0.5–1.5 mm @ 25–40 mm/s.
 - Keep retractions minimal to avoid clogging.
- Optional Post-Processing:
 - Anneal printed parts at 80–100 °C for 4–12 h to improve strength and heat resistance.

How to Print TPU (Thermoplastic Polyurethane) Filament

- Printer Requirements:
 - Direct drive extruder is highly recommended – TPU is flexible and can jam in Bowden tubes. 95A should be fine.
 - Use a printer with a fully constrained filament path to prevent buckling.
 - Works well on open-frame printers – no enclosure required.
- Recommended Print Settings:
 - Nozzle Temperature: 210–240°C (check your specific TPU brand)
 - Bed Temperature: 30–60°C (optional; helps first-layer adhesion)
 - Layer Height: 0.1–0.3 mm (0.2 mm is typical)
- Print Surface & Adhesion:
 - Sticks well to PEI.
 - Use a brim or skirt to help with initial adhesion and oozing control.
- Drying the Filament:
 - TPU is hygroscopic; it absorbs moisture easily.
 - Dry at 40–50°C for 12 hours before printing.
 - Check for <20% relative humidity before printing.
 - Store in a dry box or airtight container with desiccant.
- Speed & Retraction:
 - Print Speed: 15–60 mm/s (go slow to avoid issues)
 - Retraction: Minimal (0.5–2 mm @ 10–25 mm/s); too much can cause clogs or jams
 - Enable avoid crossing walls.
 - Disable coasting, combing, and z-hop for more consistent extrusion

How to Remove Supports from 3D Prints

- Let the Print Cool First:
 - Allow the print to cool fully before handling—supports are easier to break off cleanly when the material is rigid.
- Start with Your Hands:
 - Gently break away large support structures by hand.
 - Flex the part slightly (if it's not brittle) to loosen supports from internal cavities.
 - Be sure to support small features when removing supports.
- Use the Right Tools:
 - Flush cutters or diagonal pliers for precise nips
 - Deburring tool or X-Acto knife for edge cleanup
 - Needle-nose pliers for tight or internal areas
 - Tweezers for stringy supports in fine details
- Use Support Interface Layers (Slicer Tip):
 - When slicing, enable support interface layers (denser contact layers) for easier, cleaner breakaway.
 - Consider "air gap" settings for easier removal—typically 0.2 mm for PLA, more for PETG or TPU.
- Print Orientation Matters:
 - Orient parts to minimize supports or make them easier to remove (e.g., flat surfaces facing down).
- Stay Safe:
 - Wear gloves and eye protection when using sharp tools.
 - Cut away from your body—many support injuries come from rushed removal.

